

BIOL365: Marine and Terrestrial Ecology

Practical 4: Macroevolutionary questions

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Note from last week

- Subspecies uploaded to Genbank
 - Uploaded code under W3 — “4. Align sequences” to remove individuals by name.
 - Useful for your assignment, no?
- Change to *phangorn* package
 - `phangorn::plotBS`
 - “Bstrees” > became > “trees”
 - Function to add node support was split...
 - `phangorn::add_support`
- Changes are normal, but it’s good to be able to deal with them!
 - Probably comes with years of experience, which you are now starting to get!

What to expect today

1. Script preparation
2. Data management
 1. Genetic data
 2. Trait data
 3. Align the two
3. Phylogenetically independent contrasts
4. Ancestral state reconstructions



Script preparation

- Usually, you will do this every time you open a project
- I feel like people are getting more comfortable here



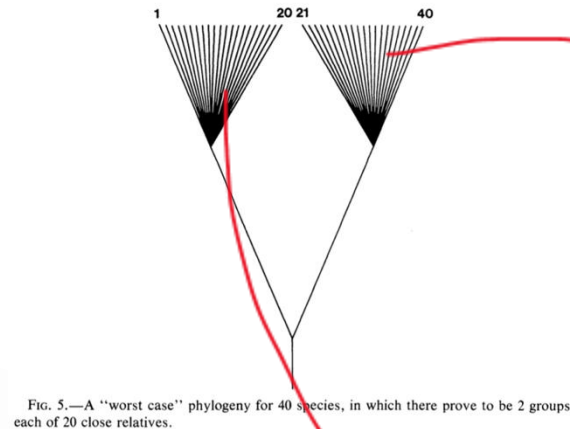
Data management

- Using the **fasta alignment** from last week
 - A bit more data management magic
 - Re-make our tree
- Read in our **trait data!**
 - Make some summaries by finch species
- Combine the two datasets
 - Some VERY useful (and pretty simple) code here



Phylogenetically independent contrasts

- Regular regression analysis
- Compare with Phylogenetically independent contrasts (PICs)



THE AMERICAN NATURALIST

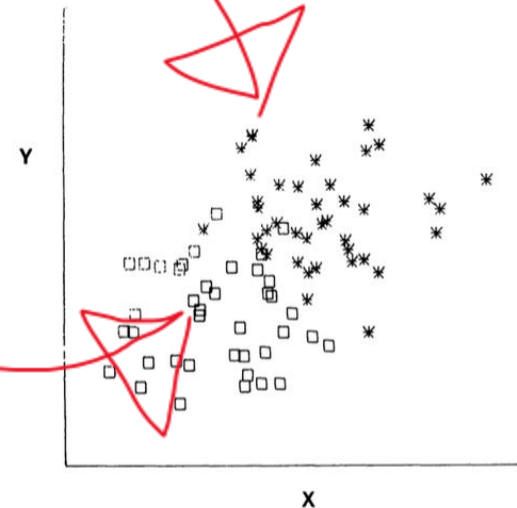


FIG. 7.—The same data set, with the points distinguished to show the members of the monophyletic taxa. It can immediately be seen that the apparently significant relationship in fig. 6 is illusory.



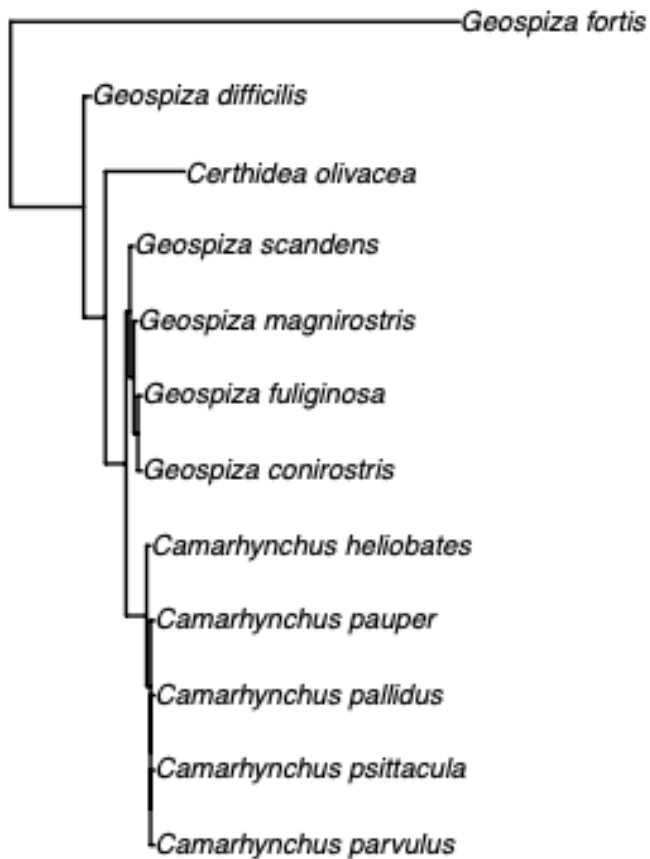
Ancestral states

- USE the tree (genetic data) and on that tree...
 - Infer how a trait has changed
 - We are not really identifying parts of the gene[ome] associated with traits

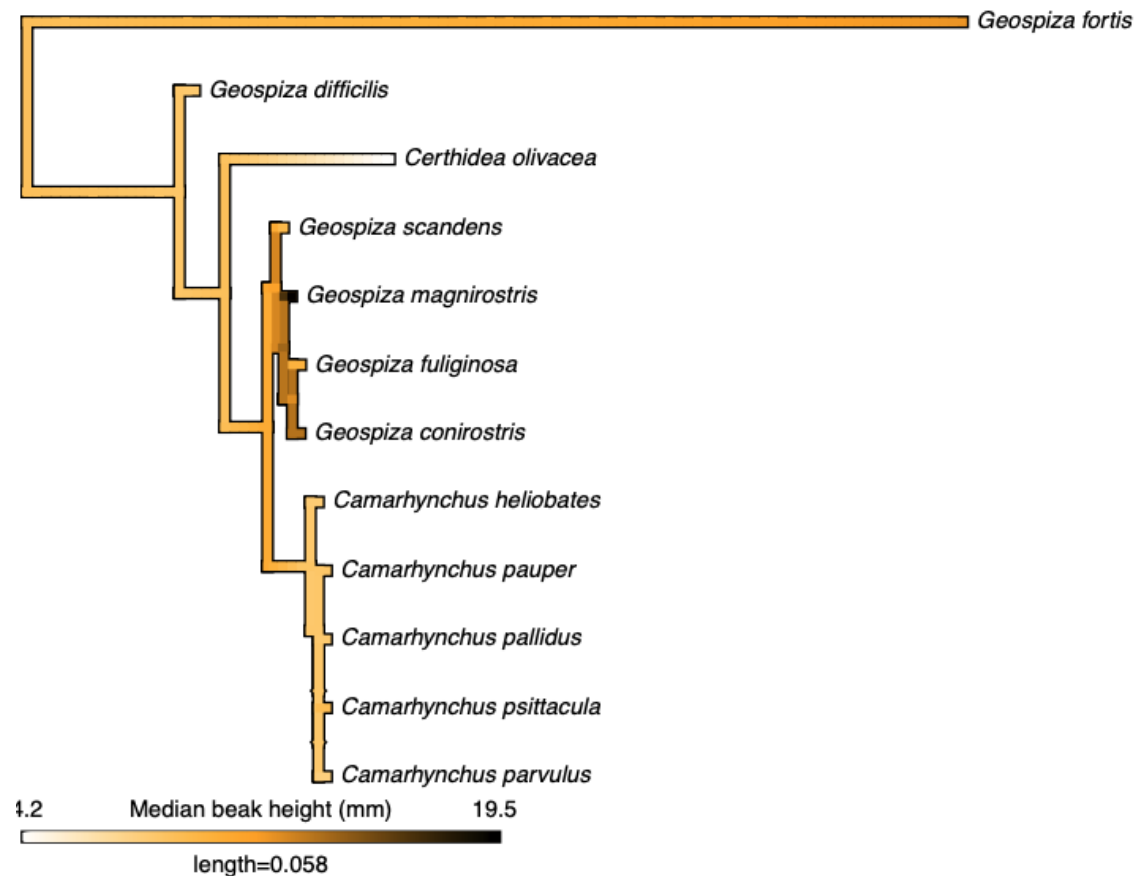


Make a phylogeny

Last week:

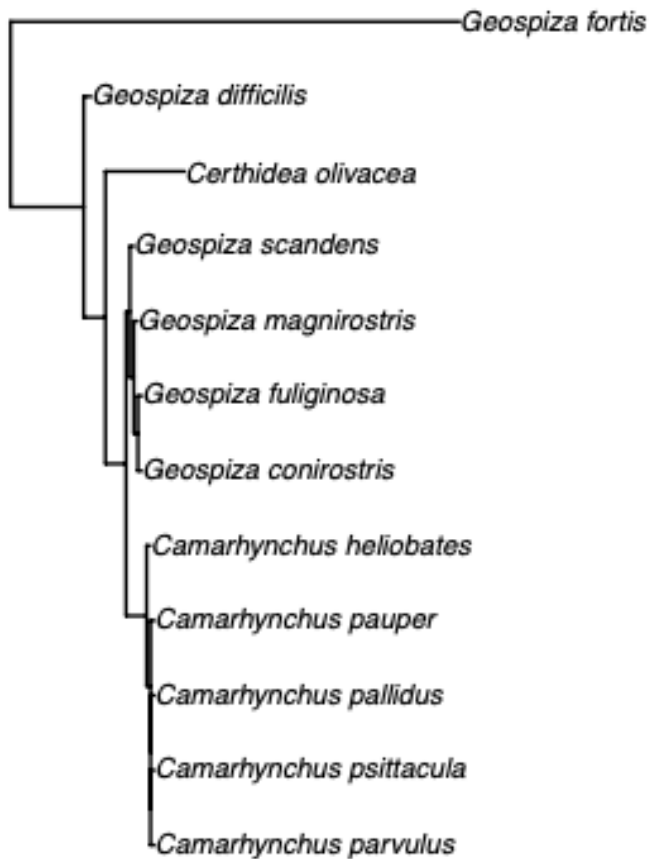


This week:

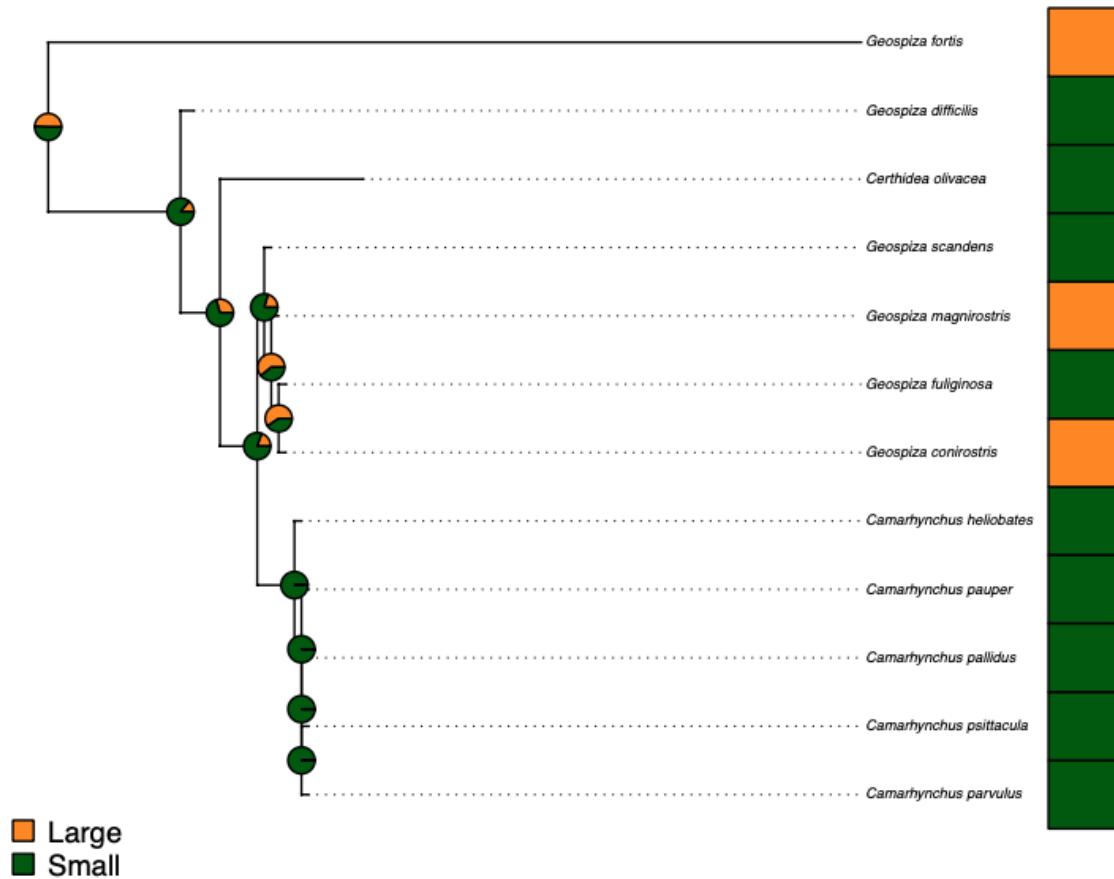


Make a phylogeny

Last week:



This week:



Phylogenetically independent contrasts

- Play with the code
- Remember, you will need to show independence and problem solving for your assignment!
- 10% component due midnight Friday after next (Week 6)



Did you finish early?

1. Help your neighbour
2. Work on your assignment and seek help or advice (keeping in mind that others doing the prac might need more help)
3. ???
4. Okay, fine you can go if you've checked with your demonstrator

